

# Dual-Domain MMD Adaptation: Small-Anchor Runs, Per-Class Breakdown

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## 1 Setup

Both directions of adaptation were run from a small (1600-image), stride-subsampled pretrained anchor checkpoint, adapted against the full opposite-domain large target set:

- **real\_small\_source\_syn\_large\_target:** source = real\_small (pretrained, detached, forward-only reference), target = syn\_large (fine-tuned via detection loss, pulled toward source via MMD).
- **syn\_small\_source\_real\_large\_target:** mirror direction, source = syn\_small, target = real\_large.

Both adapted checkpoints, plus their originating small-anchor pretrain baselines (real\_small, syn\_small), were evaluated per-class (person, vehicle) on both domains' validation sets: the synthetic val set (640 images) and the real val set (548 images).

## 2 Results

| Model                                        | person mAP50 | person mAP50-95 | vehicle mAP50 | vehicle mAP50-95 |
|----------------------------------------------|--------------|-----------------|---------------|------------------|
| real_small (pretrain baseline)               | 0.0745       | 0.0194          | 0.162         | 0.023            |
| syn_small (pretrain baseline)                | 0.679        | 0.341           | 0.693         | 0.353            |
| real_small_source_syn_large_target (adapted) | 0.687        | 0.353           | 0.696         | 0.353            |
| syn_small_source_real_large_target (adapted) | 0.125        | 0.0319          | 0.233         | 0.0319           |

Table 1: Validated on the synthetic (syn) val set.

| Model                                        | person mAP50 | person mAP50-95 | vehicle mAP50 | vehicle mAP50-95 |
|----------------------------------------------|--------------|-----------------|---------------|------------------|
| real_small (pretrain baseline)               | 0.665        | 0.311           | 0.884         | 0.311            |
| syn_small (pretrain baseline)                | 0.0125       | 0.00286         | 0.293         | 0.00286          |
| real_small_source_syn_large_target (adapted) | 0.0247       | 0.00628         | 0.318         | 0.00628          |
| syn_small_source_real_large_target (adapted) | 0.647        | 0.312           | 0.889         | 0.312            |

Table 2: Validated on the real val set.

### 3 Discussion

**Vehicle detection improves consistently in both directions.** Comparing each adapted model to its own target domain’s in-domain baseline: `real_small_source_syn_large_target` raises vehicle mAP50 on syn from 0.693 to 0.696 (mAP50-95:  $0.473 \rightarrow 0.482$ ), and `syn_small_source_real_large_target` raises vehicle mAP50 on real from 0.884 to 0.889 (mAP50-95:  $0.633 \rightarrow 0.640$ ). Both directions, both metrics, same sign – the one class that shows a genuine, if modest (0.3–0.9 point), improvement from MMD alignment in this setup.

**Person detection is inconsistent.** It improves in the syn-target direction ( $0.679 \rightarrow 0.687$  mAP50) but slightly regresses in the real-target direction ( $0.665 \rightarrow 0.647$  mAP50).

**Source-domain forgetting is present and worse for person than vehicle.** Neither adapted model receives a detection-loss gradient on its own source domain after adaptation starts. `real_small_source_syn_large_target`’s person AP on real collapses from 0.665 (as `real_small`’s own in-domain score) to 0.0247, while vehicle only drops to 0.318 from 0.884 – a much smaller relative loss. The same asymmetry holds in the mirror direction.

**Caveat.** These are single runs on val sets of modest size (160–640 images); the vehicle-specific gain, while consistent across both directions, is small enough that confirming it isn’t a favorable-seed artifact would need a repeat run (or the bandwidth-freeze + MMD-weight-decay regularization discussed in `experiment_summary_v2.tex`) before treating it as a settled result.